Title: The Use of Artificial Intelligence in Healthcare Knowledge Management

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ISM 6255 – Knowledge Management & Business Intelligence

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June 11, 2024

Spring 2024

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Abstract

This research project investigates the assimilation of artificial intelligence (AI) into healthcare knowledge management (KM), with a particular focus on the noteworthy progressions and implementations in predictive analytics, diagnostic assistance, and customized treatment planning. The conversation between AI technology and knowledge management (KM) concepts is intended to be balanced, with an emphasis on how AI may improve the generation, exchange, and use of healthcare information. The difficulties and moral issues of AI in healthcare KM are also covered in this paper.

Introduction – Knowledge Management

Healthcare knowledge management (KM) is the methodical process of generating, disseminating, using, and overseeing the knowledge and data inside healthcare institutions. Enhancing decision-making, increasing patient outcomes, and optimizing resource utilization are just a few of the ways that integrating AI into KM processes has potential to transform healthcare delivery. With an emphasis on personalized treatment planning, diagnostic assistance, and predictive analytics, this study attempts to investigate the uses of AI in healthcare knowledge management.

Introduction – Artificial Intelligence

In 1956, John McCarthy of the Massachusetts Institute of Technology proposed the idea of "artificial intelligence" as the topic of a symposium at Dartmouth. The conference's objectives were to compile findings and step-up future research on "thinking machines." The conference proved to be a huge success, leading to a significant increase in the amount of work being done on artificial intelligence-related initiatives. It is impossible to define artificial intelligence exactly. Literature has a wide range of definitions for artificial intelligence. However, it is important to note that John McCarthy provided the original definition of artificial intelligence, which is "the construction of machines that can be said to be similar to the human manifestations of intelligence."

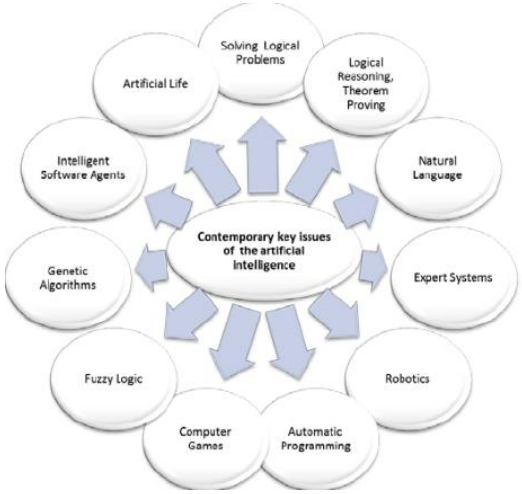

Fig. 1 Contemporary key issues of artificial intelligence

Knowledge Management in Healthcare

Definition and Importance

Healthcare knowledge management (KM) refers to the procedures used to gather, share, and use information efficiently. It includes procedures including the generation, archiving, retrieval, and distribution of knowledge. Due to the complexity and quick development of medical knowledge, the necessity of evidence-based practice, and the significance of multidisciplinary collaboration, effective knowledge management is essential in the healthcare industry (Davenport & Prusak, 1998).

The elements of knowledge management

Healthcare's key knowledge management components are as follows:

1. According to Davenport and Prusak (1998), knowledge creation is the process of producing new understandings and data via research and clinical application.
2. Storing and retrieving knowledge in an organized and usable manner is known as knowledge retrieval and storage (Davenport & Prusak, 1998).
3. Knowledge sharing: According to Davenport and Prusak (1998), this is the process of sharing knowledge inside the company to guarantee full utilization.
4. According to Davenport and Prusak (1998), knowledge application is the process of applying knowledge to guide clinical judgement and enhance patient care.

Table 1. Potential AI applications in different KM processes

The KM process	Possibilities created using AI systems	Examples of use cases
Knowledge creation	• Fostering predictive analytics via self-learning analytical capacities • Recognizing previously unknown patterns • Sifting through organizational data and discovering relationships • Developing new declarative knowledge	• Forecast sales probabilities • Discover organization inefficiencies by analyzing CRM records
Knowledge storing and retrieving	• Harvesting, classifying, organizing, storing, and retrieving explicit knowledge • Analyzing and filtering multiple channels of content and communication • Facilitating knowledge reuse by teams and individuals	• Organize and summarize legal precedents relevant to a new case • Retrieve dispersed nuggets of information related to a troubleshooting situation
Knowledge sharing	• Connecting people working on the same issues by fostering weak ties and know-who • Facilitating collaborative intelligence and shared organizational memory • Generating a comprehensive perspective on knowledge sources and bottlenecks • Creating more coordinated, connected systems across organizational silos	• Facilitate feedback and peer review on communication systems (e.g., Slack) • Facilitate real-time smart sharing between marketing channels and sales pipelines

Artificial Intelligence in Healthcare

Definition and Capabilities

AI is used in healthcare to describe a range of technologies that can handle massive quantities of data, find patterns, and make predictions. Examples of these technologies are robots, machine learning (ML), and natural language processing (NLP). Through better data analysis, easier information retrieval, and more accurate decision-making, artificial intelligence (AI) can improve knowledge management (KM) (McKinsey Global Institute, 2018).

Predictive Analytics

Using past data to forecast future occurrences with confidence is known as predictive analytics. AI-driven predictive analytics in healthcare can identify at-risk groups, anticipate disease outbreaks, and forecast patient outcomes. By analyzing electronic health records (EHRs), for instance, AI algorithms can identify individuals who are most likely to acquire chronic diseases, allowing for early intervention and preventative therapy (Obermeyer & Emanuel, 2016).

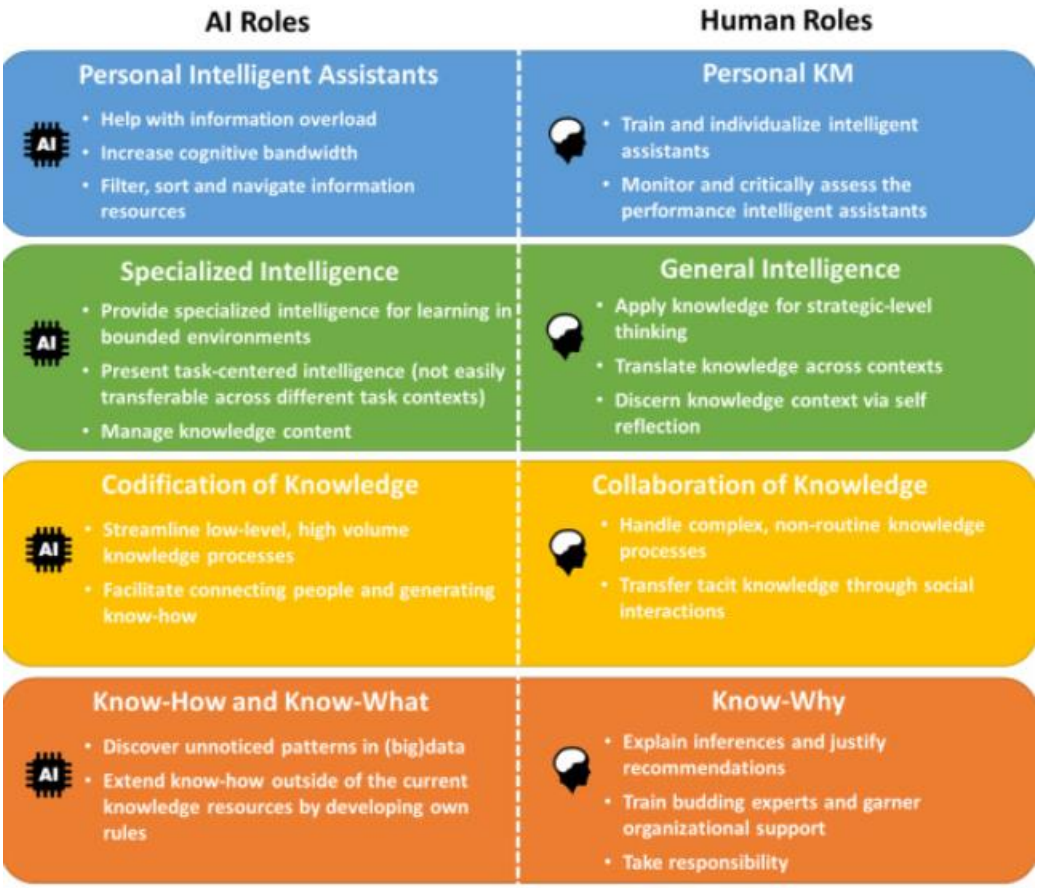

Fig2. A symbiotic relationship between humans and AI in managing knowledge

Diagnostic Support

By examining medical pictures, test findings, and other diagnostic data, AI can help with illness diagnosis. To help medical personnel diagnose patients accurately, machine learning models can recognise patterns and abnormalities that may lead to certain illnesses. Topol (2019) reports that artificial intelligence (AI) has demonstrated a high degree of accuracy in detecting cancer in radiological pictures, which might raise the rate of early diagnosis.

Personalized Treatment Planning

The process of creating a patient's personalized course of care includes addressing each patient's unique medical needs. With AI, individualized care plans may be created by analyzing lifestyle, genetic, and therapeutic data. By tailoring medicines to the individual needs of each patient, this method can increase treatment success and decrease side effects (Topol, 2019).

Practical implications

A recent survey indicates that a vast majority of AI investments bring about minimal or no impact (Ramsbotham, Khodabandeh, Fehling, et al., 2020). Years of research indicate that for an IT deployment to be successful, there needs to be accompanying organizational changes. These are referred to as organizational complements (Brynjolfsson et al., 2017). So, the value of AI for KM lies not only in technology but in new infrastructures, trained people, and redesigned processes. We discuss the practical implications of AI and its role in KM based on the three complements of people, infrastructure, and process

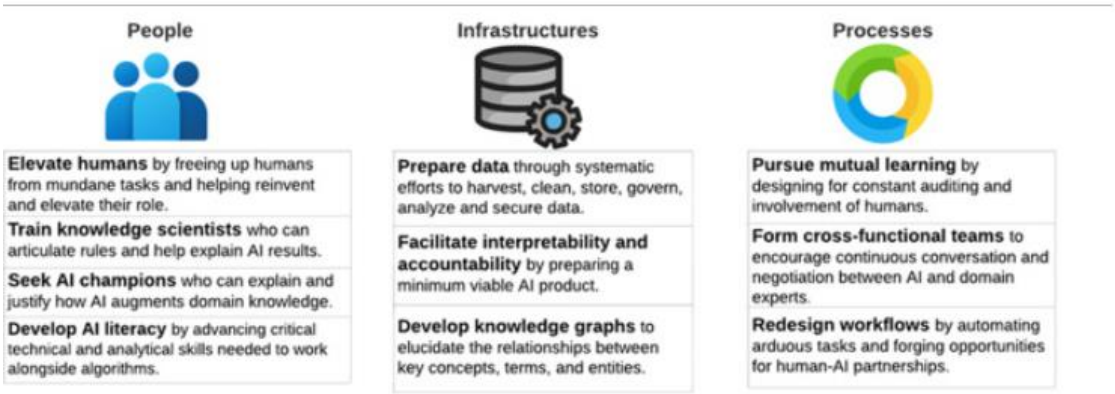

Fig3. Organizational changes needed to accompany AI adoptions for KM

AI in Healthcare Knowledge Management

Enhancing Knowledge Creation

AI can speed up the development of new knowledge by sifting through large information and finding novel patterns or insights. AI has the ability to evaluate and understand clinical trial data, for instance, revealing novel connections and possible treatment approaches that would not be seen using more conventional research techniques (Wu et al., 2017).

Improving Knowledge Storage and Retrieval

Artificial intelligence (AI) technologies, such natural language processing (NLP), can improve knowledge retrieval and storage by converting unstructured material into organized representations that facilitate search and access. According to Wang, Kung, and Byrd (2018), AI-powered systems can also give healthcare professionals pertinent information at the time of treatment, enhancing their ability to make decisions and improve patient outcomes.

Facilitating Knowledge Sharing

AI can support knowledge sharing by connecting healthcare professionals through intelligent platforms that recommend relevant research, clinical guidelines, and expert opinions based on their needs and preferences. These platforms can foster collaboration and ensure that the latest knowledge is disseminated throughout the organization (Raghupathi & Raghupathi, 2014).

Enhancing Knowledge Application

AI can assist healthcare professionals in applying knowledge by providing decision support tools that integrate clinical guidelines, patient data, and evidence-based practices. These tools can offer recommendations and highlight potential risks, helping clinicians make informed decisions and deliver high-quality care (Bishop, 2006).

Challenges and Ethical Considerations

Data Privacy and Security

The use of AI in healthcare KM involves handling sensitive patient data, raising concerns about privacy and security. Ensuring that AI systems comply with regulations such as the Health Insurance Portability and Accountability Act (HIPAA) is essential to protect patient information (Obermeyer & Emanuel, 2016).

Bias and Fairness

AI algorithms can perpetuate existing biases in healthcare data, leading to unfair treatment and disparities. It is crucial to develop and implement AI systems that are transparent, unbiased, and

equitable, ensuring that all patients receive fair and appropriate care (McKinsey Global Institute, 2018).

Ethical Implications

Concerns about openness, accountability, and autonomy are raised by the use of AI in healthcare knowledge management. It is imperative for healthcare institutions to institute ethical norms and supervision systems in order to tackle these concerns and guarantee the responsible and ethical application of AI (Topol, 2019).

Individual K.M. and personal intelligent assistants

Organizational knowledge management (KM) incorporates a significant personalization component that centers on individual employees and their "personal inquiry" (Pauleen & Gorman, 2016). Because of this, conceptions of knowledge frequently acknowledge the importance of each worker in defining the notion of knowledge. Knowledge is defined as "a fluid mix of framed experience, values, contextual information, and expert insight that provides a framework for evaluating and incorporating new experiences and information.

Over time, personal intelligent assistants are interactive, self-learning systems that may improve and adjust how they engage with the user. By doing this, they may discover the changing knowledge requirements, processes, preferences, and feedback of each employee and offer tailored solutions. For instance, personal intelligent assistants are used with traders at one financial institution to help manage information overload and increase employees' cognitive capacity (Ransbotham, Khodabandeh, Fehling, et al., 2020). These personal intelligent assistants are trained to aid traders by bringing pertinent information to their attention in real time after identifying

trends in the traders' data-monitoring procedures, decision-making, and trading outcomes. In its capacity as the traders' personal intelligent assistant, this timely information helps them make better judgements and symbolises a partnership that is always evolving.

People In Knowledge Management, humans are elevated

The crucial role that workers may play in AI-powered automation is not well conveyed by metaphors like "human-in-the-loop" AI or "the last mile problem." Humans are frequently portrayed in these formulations as an afterthought, an adjunct to the technical AI system, or a barrier to complete automation. Rather, we propose a symbiotic connection that looks for methods to both elevate and reimagine the role of people while also acknowledging their indispensable contributions to knowledge work. The personnel at the front lines of organizations that incorporate AI into their work habits are the most crucial to the success of AI.

Educate future knowledge scientists

According to Doran et al. (2017), recent advancements in explainable AI (XAI) and neural-symbolic learning support the creation of new positions such as knowledge scientists and data scientists, who gather and prepare training data sets for machine learning algorithms. Building knowledge graphs that complement training data and reflect background knowledge is one way that knowledge scientists may assist integrate the two separate AI strategies: statistical AI (based on neural networks) and symbolic AI (using more conventional techniques). By expressing rules and related explanations, knowledge graphs may be utilized to improve the transparency and explainability of decisions made by the AI agent. Thus, knowledge scientists and machine learning

algorithms collaborate and are essential components of this strategy for explainable artificial intelligence (Blumauer, 2019).

Look for champions of AI

The presentation of an alternative narrative that stresses supplementing knowledge workers rather than replacing them—a narrative that outlines the anticipated improvement in the types of activities knowledge workers perform—can be greatly aided by advocates of artificial intelligence. Boundary spanners, or AI champions, identify connections between certain KM needs and functions and AI capabilities. These are people who understand and are enthusiastic about artificial intelligence (AI), but more significantly, they have strong commercial and communication acumens, in addition to deep topic expertise, which enables them to articulate how AI can enhance people's abilities and knowledge processes.

Conclusion

There is a lot of opportunity to improve knowledge production, storage, retrieval, sharing, and application by integrating AI into healthcare knowledge management. Healthcare companies may enhance patient outcomes and maximize resource utilization by utilizing AI technologies like personalized treatment planning, diagnostic assistance, and predictive analytics. To fully reap the benefits of AI in healthcare knowledge management, however, issues with data privacy, prejudice, and ethical considerations must be adequately addressed. The development of strong, open, and fair AI systems that improve healthcare knowledge management while defending patient rights and advancing moral principles should be the main goal of future research and policy initiatives.

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